INTRODUCTION

This document outlines basic procedures for installing the MS7001F/MS7001FA Gas Turbine Load Coupling using the Tension Stud and Nut Tooling provided by Riverhawk Company. Specific information for installing these bolts on other turbine models can be found at the end of this document in Exhibit "A", Sections 1 through 4. These guidelines apply to a typical centerline installation, and do not supersede any factory instructions.

EQUIPMENT DESIGN

The Gas Turbine Load Coupling is designed to be installed in the field after the turbine has been set down on the foundation, leveled, aligned to foundation centerlines, and locked into place. A set of coupling bolts must be assembled to the load coupling–turbine end.

The coupling will then be used as an alignment fixture to align the generator to the turbine. When final alignment is completed, the set of coupling bolts will be assembled to the coupling on the generator end.

HEALTH AND SAFETY REQUIREMENTS

In all cases, the Health, Safety and Environmental Manual (HSEM) will be the document followed on the job site during installation of this equipment.

Safety, health, and environmental protection must be an integral part of all department business operations, including design, construction and start-up. Every reasonable effort will be made to create and maintain a safe and healthful work environment at each customer site.

REFERENCE DOCUMENTS

No.	Description
MLI 0411	Basic Field Alignment Instructions
IM115 Pub.	Riverhawk – Instruction Manual on Coupling Tension Studs and Nuts

EQUIPMENT ABBREVIATIONS

The documents referenced in this article are identified by an equipment group prefix: “MLI” (Model List Item) indicates Gas Turbine materials, “GEN” indicates Generator materials, “BOP” indicates Balance of Plant materials, and “MSD” indicates Material Shipped Direct by vendor.

1.0 GENERAL INFORMATION

This document describes the procedure to be used when installing studs and nuts (supplied by Riverhawk Company) in the flanges at the turbine/coupling end and the coupling/generator end connections. This hardware tooling is depicted on drawings included throughout this document.

2.0 ASSEMBLY PROCEDURE

2.1 Machine Preparation

The flanges to be tensioned must be fully closed prior to positioning the studs in the flanges, and there must be provision to turn the turbine and generator shafts. It will also be advantageous to remove as many obstructions as possible from the flange area (such as speed probes and conduit).

2.2 Hardware – Balance

- Hardware is supplied as weight-balanced sets.
- Studs and Nuts are interchangeable only within sets.
- Do not mix different sets together.
- Retain the weight certification supplied with each set for the purchase of spares.

2.3 Tensioner – Care and Handling

- When not in use, the tensioner must be maintained in a clean environment. All caps, plugs for hydraulic openings, and fittings must be in place.
- When in use, the tensioner must be protected from sand and grit.
- For long term storage – coat the tensioner with oil, return to its original container, and seal container to protect from moisture.
- For shipment, coat the tensioner with oil and ship in its original container.

2.4 Hand Tools

Several hand wrenches and micrometers will be needed to install and measure the studs.

MS7001/MS7001FA Gas Turbines	
5/8 in. Wrench	5/16 in. Hex Key Wrench
1.0 in. Wrench	2.75 in. Wrench
1.0 in. Socket	9 in. – 10 in. Micrometer or Caliper
1/2 in. Hex Key Wrench	10 in. – 11 in. Micrometer or Caliper

2.5 Special Tools

- Hydraulic Pressure Kit MP–0130 (hand pump), or AP–0532 (air-driven pump recommended).
- Hydraulic Tensioner HT–0342.

3.0 HARDWARE PREPARATION

3.1 Nut Preparation

- For new installations, the nuts should arrive sealed from the factory with no need for extensive cleaning.
- Nuts that have been used in a previous installation will require cleaning. Clean the nuts with a wire brush and a petroleum based solvent to remove any foreign material on the external surfaces and threads.
- If the previous installation employed a thread locking compound (which will be visible as a grayish-green residue), remove as much of this compound as possible.

  Note: Do not apply lubricants to the threads.

- Finish cleaning the nuts by rinsing in a volatile solvent (such as acetone) and allow to dry.

3.2 Stud Preparation

- For new installations , the studs should arrive sealed from the factory with no need for extensive cleaning.

- Studs that have been used in a previous installation will require cleaning. Clean the studs with a wire brush and a petroleum based solvent to remove any foreign material on the shank and threads.

- If the previous installation employed a thread locking compound (which will be visible as a grayish-green residue when the nut is removed), remove as much of this compound as possible.

  Note: Do not apply lubricants to threads.

- Finish cleaning the studs by rinsing in a volatile solvent (such as acetone) and allow to dry.

3.3 Stud Length Measurement

Measure and record the initial lengths of the studs. Following the recommendations below will improve the results.

- Plan to start and finish the flange in the same day.

- Studs and flange must be at equal temperature.

- Number each stud with a marker.

- Mark the measurement location on the stud end with a permanent marker.

- Measure each stud to nearest 0.001 inch.

- Record each measurement on the supplied charts.

- Do not expose the measuring instruments to the sun for any length of time.

4.0 STUD AND NUT ASSEMBLY

Refer to hardware assembly drawing for configuration. Assemble cylindrical nut to the tapered thread end of the stud. Slide the assembly into the flange from the coupling side as shown in Figures 1 and 2. Install the hex nut on the backside. Adjust the assembly so that the stud protrudes from the cylindrical nut as depicted on the hardware drawing.

This is also important for proper assembly of the windage plates. Refer to Section 11.0. Hand tighten the assembly to a snug fit. See Figures 3 and 4 for a view of studs in flange prior to tensioning.

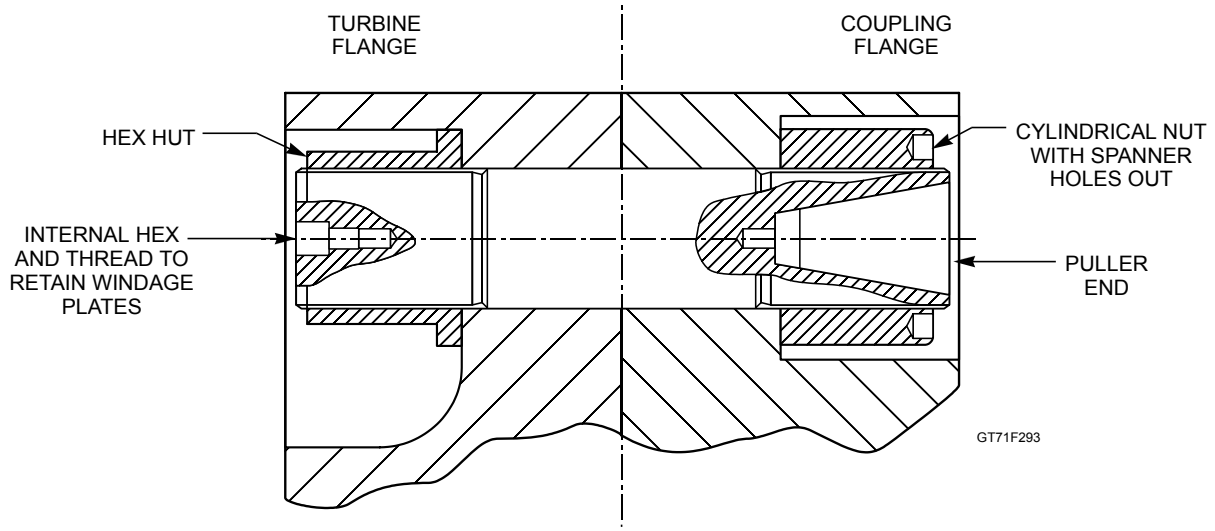


Figure 1. Stud Assembly Without Windage Plates

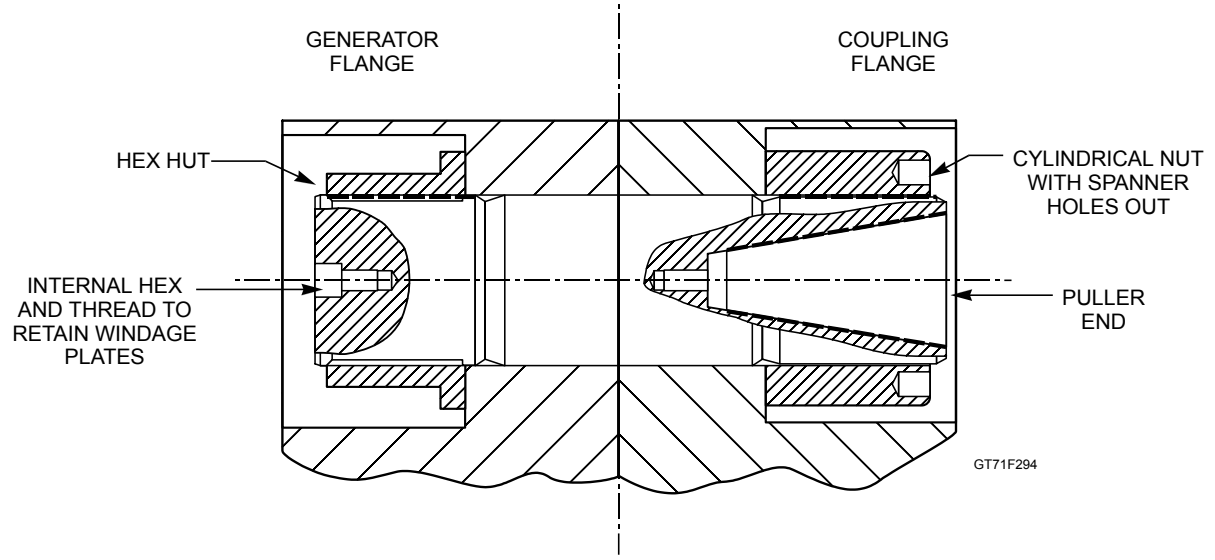


Figure 2. Stud Assembly Without Windage Plates



Figure 3. Section of MS9001FA Turbine Flange with Studs in Place Prior to Tensioning – Typical Layout

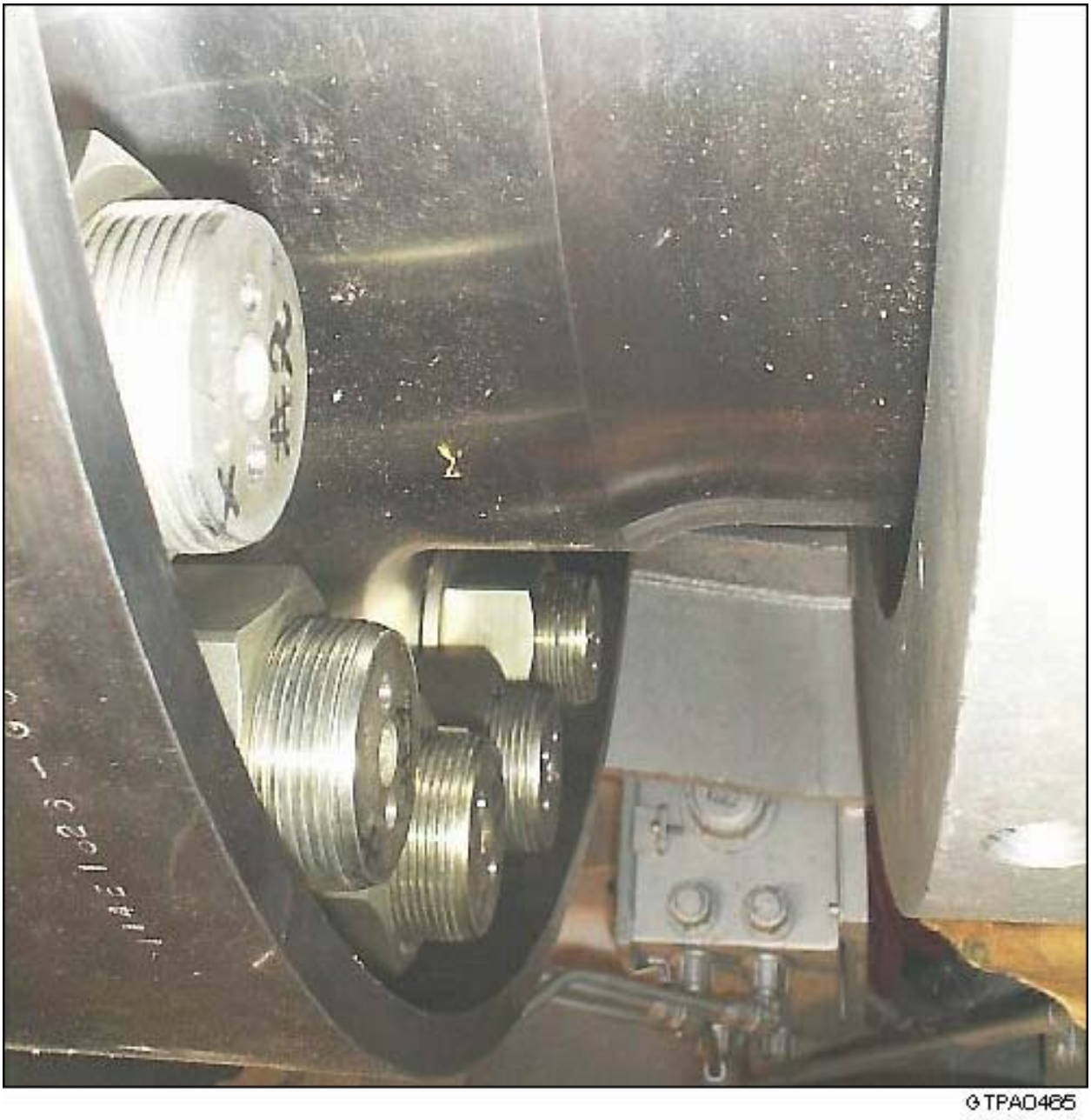


Figure 4. Compressor Side of MS9001FA Turbine Flange with Studs and Nuts in Place – Typical Layout

5.0 ASSEMBLING HYDRAULIC TENSIONER EQUIPMENT

5.1 Kit Assembly

Assemble the hydraulic pump with its hose to the puller tool and bleed the system of air per the following instructions. Figure 5 shows hose connected to tensioner manifold.

5.1.1 Fittings

Ensure that both the male and female parts are clean and free of debris (see Figure 6 for fitting configuration). Hold female part securely when tightening to prevent damage to adjacent tubing. If fitting leaks, retighten as needed. If leaking continues, disassemble and check for scratches or debris on the seating conical surfaces. Clean as necessary. Replace the plastic caps when finished with tooling.

5.2 Pump

The pump kit is filled with hydraulic oil when shipped, and the pump reservoir cap sealed for containment. To prevent oil spillage, keep cap closed when not in use, or during storage and shipping. Lost oil should be replaced with Enerpac Hydraulic Oil, ISO 32. Mineral oil may be substituted if necessary.

5.3 Bleeding the Hydraulic System

Bleeding air from the hydraulic system must be performed before stud tensioning begins. Follow the tensioner assembly instructions in Section 6.0.

The tensioner has four ports (see Figure 7); one for pressurizing, two for bleeding the system, and one for pressure relief. To facilitate bleeding, start by mounting the tensioner at either the 3 o'clock or 9 o'clock stud position, depending on which will critically place the bleed ports in their uppermost position. In addition, ensure that the pump is always situated below the tensioner assembly.

The puller tool is equipped with 5/8 in. Hex cone stem bleeder fittings installed in the bleeder ports. With these two loosened simultaneously, stroke the pump repeatedly until the streams of oil exiting the tool from each port are free of air. Retighten the fittings.

Note: The hose is stiff. Use of this tooling can be simplified by temporarily mounting the puller tool on one stud prior to final tightening of the fittings. This will reduce the tendency for the fittings to loosen during use.



Figure 5. Tensioner Tool in Operation While Mounted on MS9001FA Turbine Flange

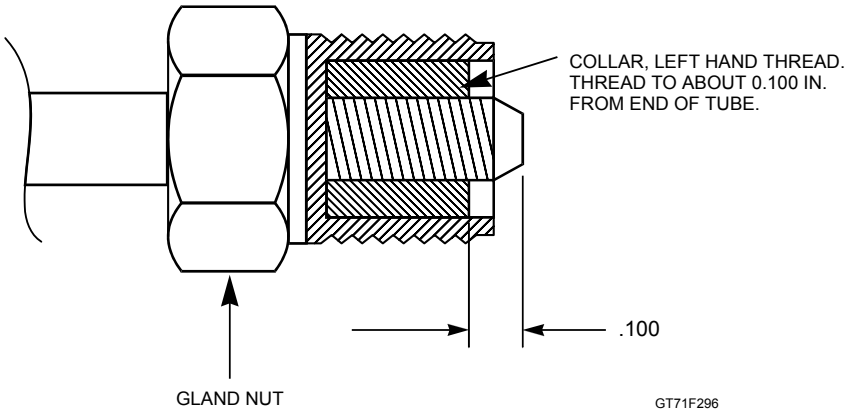


Figure 6. Pressure Port

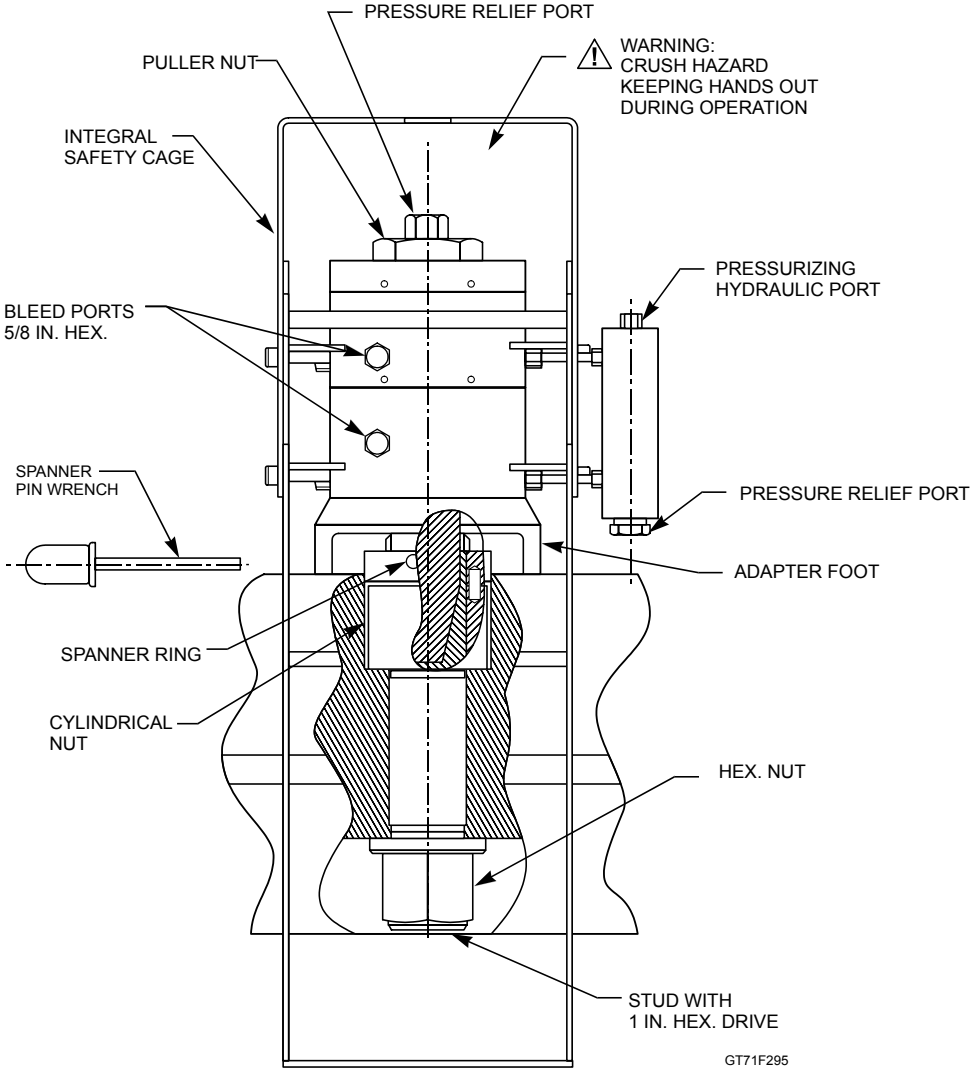


Figure 7. Bolt Tensioner Tool

6.0 ASSEMBLY OF TENSIONER ON STUD

Refer to the Tensioner Assembly drawing and Figure 7 for views of tensioner to flange mounting. All tensioning is to be performed on the coupling side of the flange connection.

Note: The HT–0342 is supplied with two adapter feet and two spanner rings (the feet are marked “turbine” and “generator”). Mount the appropriate foot before attempting to use the tensioner.

6.1 The following features should make stud tensioning safe and easy.

- The safety cage is integral (bolted) to the puller.
- The hydraulic piston is spring-loaded to retract.
- The two-piece puller screw requires the operator to tighten the puller screw into the stud.

6.2 Assemble as follows.

1. Place the spanner ring on the cylindrical nut. **Ensure that the puller nut is loose at the start of each assembly.**
2. Place and hold the tensioner assembly over the end of the stud to be tightened. (See Figures 5, 8 and 9.)
3. Slide the puller screw into the tapered thread of the stud and hand tighten. **Ensure assembly is not crossthreaded.**
4. Hold the stud steady with a 1/2 in. hex key wrench, and lightly tighten the puller screw into the thread with a wrench.
5. Turn the puller nut until it seats snuggly, then back off 2 flats. This is particularly important for removal because the stud shortens during disassembly and the tensioner may bind.
6. At this point the Tensioner Assembly **MUST BE FREE TO ROTATE.** The puller screw must be tight in the stud and the puller nut must be backed off 2 flats.

Note: If the puller tool is not free to rotate when the puller screw is tight , then either (1) the puller nut is too tight, or (2) the stud is not properly positioned in the flange. The nuts must then be repositioned so that the stud is shifted slightly more on the puller tool side. This can be done as follows.

1. Slightly loosen the puller screw.
2. Back the nut opposite the puller tool off about 1/2 turn.
3. Tighten the puller screw side nut to take up the slack.
4. Retighten the puller screw per above and check for looseness of tool.

Note: Do not over extend the stud. Over extending can cause the piston to lose its seal and leak oil.

CAUTION: **Personal injury and equipment damage can occur if the puller screw is not securely engaged with the tapered thread of the stud. Proper engagement is achieved when the puller screw is tight in the stud and the Tensioner Assembly is free to rotate.**

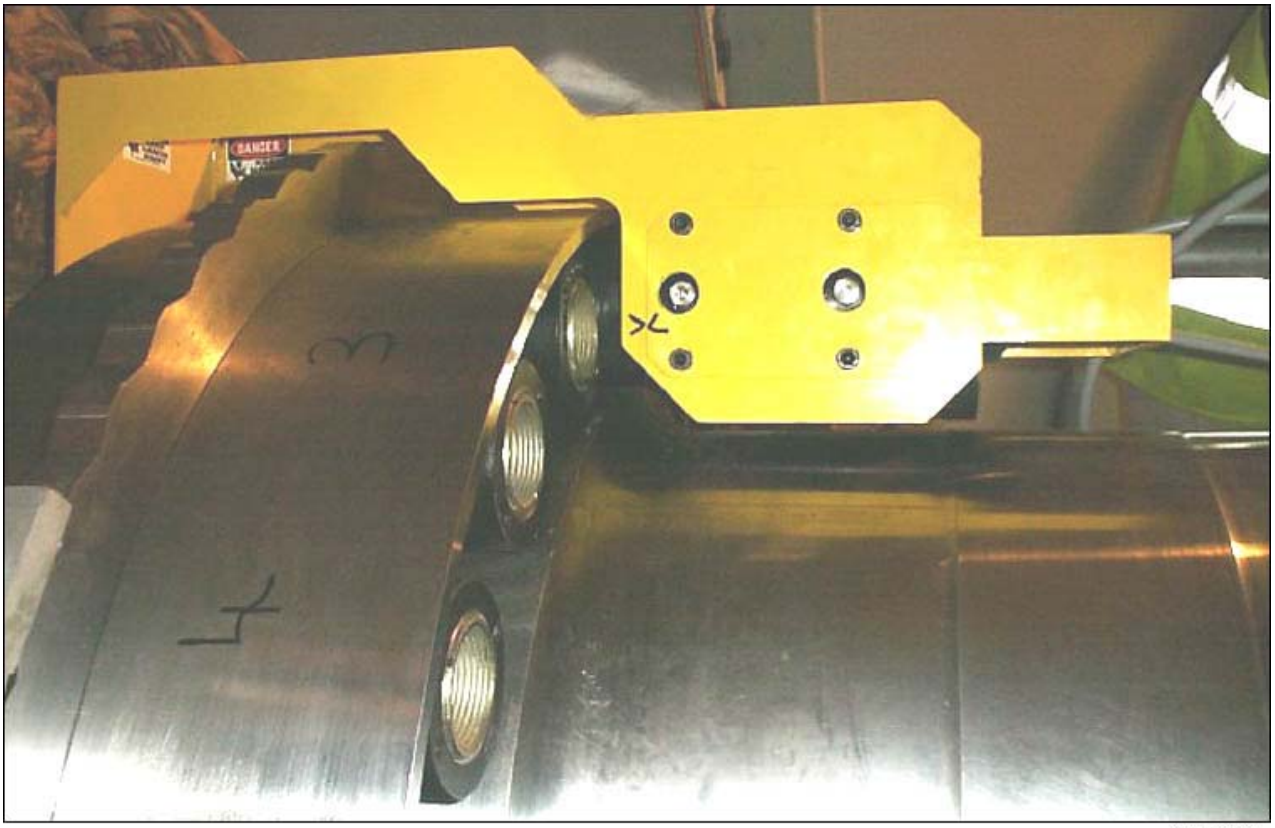


Figure 8. Side View of Tensioner Mounted on MS9001FA Turbine Flange



Figure 9. Tensioner In Operation. Note the Pressure Kit Pin Wrench and Control Handle

7.0 STUD PULLING AND TENSIONING

The studs are to be tensioned in two steps (once at approximately 50% pressure and once at final pressure). Follow the sequence for tensioning each flange joint as defined on the data sheets at the end of this instruction.

Note: Before inserting the puller screw into the tapered thread of the stud, carefully clean both the male and female threads and apply a high pressure lubricant (such as “Never Seize”) to the male taper. This will help ease assembly and ensure positive mating of the threads before tightening.

WARNING: The safety cage MUST be in place. Keep hands out of designated areas at all times when the puller tool is pressurized to prevent personal injury.

7.1 Tensioning at 50% Pressure

After the tensioner is properly installed, apply hydraulic pressure to the tool. Bring the pressure to the 50% level in accordance with the following table.

MS7001F/MS7001FA Gas Turbines		
Flange	**Stud Diameter**	**50% Pressure**
Turbine/Coupling	2.283 in.	9,000 psi
Coupling/Generator	2.784 in.	10,000 psi

7.1.1 Turn the cylindrical nut using the spanner ring and pin (as depicted in Figure 7) until it bottoms out on the flange.

7.2 Puller Tool Removal

Remove the puller tool as follows.

1. Release the puller tool pressure by opening the valve on the pump. Leave the valve open.

2. Using a wrench, unscrew the puller screw.

3. Tap the wrench with a hammer if necessary to help loosen the puller screw.

4. Move the tool to the next stud/nut assembly to be tensioned, following the sequence/pattern as defined on the supplied data sheets.

7.3 Tensioning at Final Pressure

Repeat the pulling and tightening procedure described in Sections 7.1 and 7.2 at full pressure. Measure the length of the studs after all studs have been tensioned. The final pressure and required stretch values are listed in the following table.

MS7001F/MS7001FA Gas Turbines			
Flange	**Stud Diameter**	**Stretch**	**Final Pressure**
Turbine/Coupling	2.283 in.	0.018/0.022 in.	18,000 psi
Generator/Coupling	2.784 in.	0.013/0.016 in.	19,000 psi

CAUTION: Do not exceed the pressure marked on the puller. Excessive pressure can damage the stud and the puller screw.

8.0 RETENSIONING

Excessive stretch variation or low stretch values can be corrected by retensioning all or selected studs to the pressure values stated above. Final stretch values must be approved by the owner representative, installer millwright foreman, and lead GE Technical Advisor.

9.0 THREAD LOCKING

Once pulling and tensioning is complete, all stud nuts must be locked in position. Two methods of thread locking may be encountered in the field. Early versions of the hardware entails a liquid thread-locking compound, while later configurations employ a mechanical locking feature. Each method is described in detail in Sections 9.1 and 9.2.

9.1 Thread Locking Using a Liquid Locking Compound

These nuts have no visible locking feature. Apply the specified number of drops (see following table) of thread locking compound (Permatex Industrial “After Lock” No. 81794, included with each set of hardware from the factory) to each end of the stud at the stud/nut interfaces follows.

1. Apply the drops to the face of the nut at the top of the assembly, allowing material to run down onto the stud threads.

2. An alternate method of applying the thread-locking compound is to use a short (2 to 3 in.) length of wire. Place one end of the wire at the top of the nut/stud interface with the wire held at a 45 degree angle, and apply the liquid to the wire so that it runs along the wire into the threads.

MS701F/MS7001FA Gas Turbines		
Location	**Stud Size**	**Amount of Liquid**
Turbine End	2.283 in.	6 drops
Generator End	2.784 in.	7 drops

CAUTION: Do not use more thread locking compound than specified, or the nut will be VERY difficult to remove during disassembly.

9.2 Thread Locking Using a Mechanical Locking Device

Mechanical locknuts have two set screws located in the top face (see Figure 10). Before threading the nut onto the stud, check to ensure that the set screws rotate freely. When the nut is seated, torque the set screws to the values specified in the following table. When seated and torqued to the specified values, the load created by the set screw will displace the thread of the nut in the area of the web, thus creating the desired locking action.

MS7001F/MS7001FA Gas Turbines		
Nut Diameter (in.)	**Set Screw Size**	**Seating Torque (in./lbs.)**
2.250 in.	1/4 – 28	80 to 90
2.750 in.	3/8 – 24	200 to 250

10.0 STUD AND NUT REMOVAL

The procedures for removing nuts that have been locked with liquid locking compound and those with the mechanical locking feature are described in Sections 10.1 and 10.2, respectively.

10.1 Removal of Assemblies with Liquid Locking Compound

For assemblies that have been locked with the liquid locking compound, removal is performed as follows.

1. Using a wire brush and shop air, clean the internal tapered thread of the stud to remove any debris and deposits that may have accumulated during service.

2. Install the appropriate puller tool to the stud as described in Section 6.0.

3. Apply hydraulic pressure per the following table and, using reasonable force, loosen the nut using the spanner/pin wrench (as shown in Figure 7).

4. If the nut cannot be loosened, release the pressure and repeat the procedure.

Note: Two or three attempts are usually sufficient to break the bond.

5. If the nut cannot be loosened after three attempts, heat will need to be applied to facilitate loosening.

WARNING: **Fire Hazard. DO NOT heat when puller assembly is in place. Personal injury or equipment damage may occur. Use of an Oxy-Acetylene torch is not recommended.**

Apply a smear of 550/650 degree F tempil stick to the side of the nut opposite the application of heat, and heat the nut using a propane torch. Continue to apply heat until the tempil smear indicates that the nut has reached 550 to 650 degrees F. **Do Not overheat to a cherry red condition.** Remove the heat source as quickly as possible, reinstall and apply the appropriate puller tool, apply the appropriate pressure and loosen the nut. Then release the pressure and remove the puller tool.

MS7001F/MS7001FA Gas Turbines		
Location	**Nut Size**	**Puller Pressure**
Turbine End	2.250 in.	18,000 psi
Generator End	2.750 in.	19,000 psi

10.2 Removal of Assemblies with Mechanical Locknuts

For assemblies which have been locked using mechanical locknuts, removal is performed as follows.

1. Using a wire brush and shop air, clean the internal tapered thread of the stud to remove any debris and deposits which may have accumulated during service.

2. Loosen the two locking set screws with an Allen Wrench, but do not remove from the nut (see Figure 10).



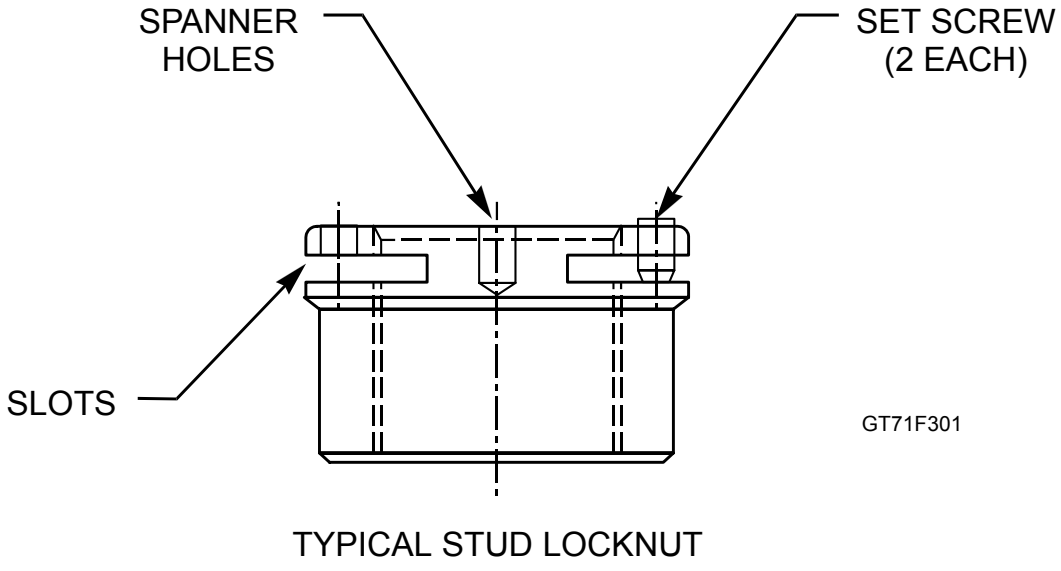


Figure 10. Generator or Turbine Ends of Stud with Locknut

3. Install the appropriate puller tool to the stud as described in Section 6.0.

4. Apply the appropriate hydraulic pressure per the Section 10.1 table and, using the spanner ring and the spanner/pin wrenches shown in Figure 7, loosen the nut. Then release the pressure and remove the puller tool.

11.0 INSTALLING WINDAGE PLATES

If windage plates are supplied, they are to be installed after the stud assembly is complete. Plate locations are shown on Assembly Drawing HF-0769, and on Figures 11 and 12. Install windage plates as follows.

1. Apply several drops of thread locking compound (packaged with the hardware) to a clean retainer screw before inserting into the stud. **Torque to 30-40 in./lbs. Do not over torque.** See Figure 13 for view of windage plates.

2. To ensure proper assembly of the windage plates, check the “before tensioning” dimension and “Note” shown on Figures 11 and 12. This will ensure that the plates are securely contained within the confines of the respective counterbores and recesses of the flanges. Refer to Figures 14 and 15 for more windage plate detail.

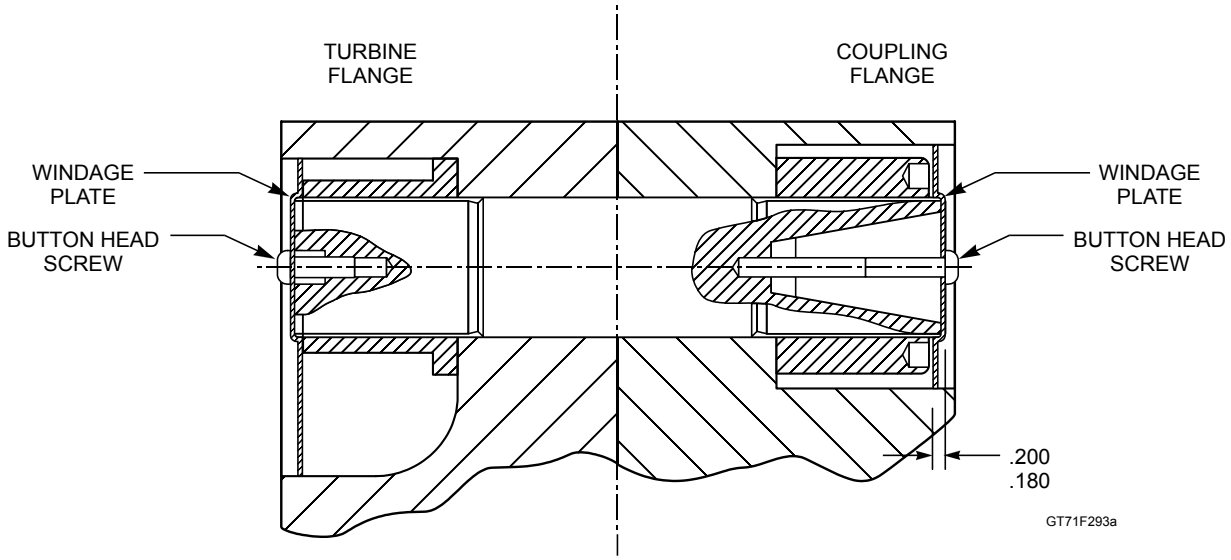


Figure 11. Stud Assembly with Windage Plates

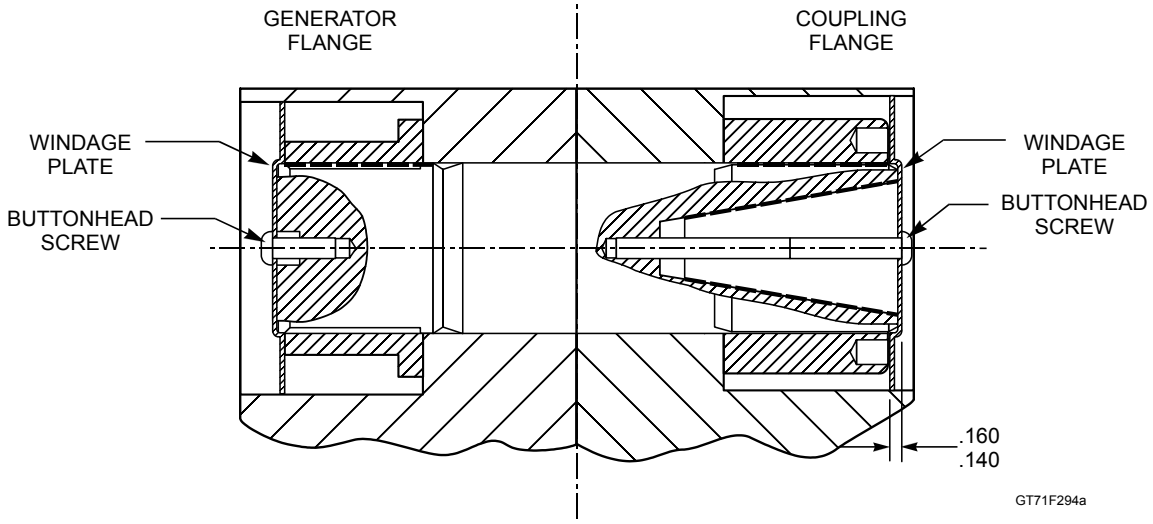


Figure 12. Stud Assembly with Windage Plates



Figure 13. View of Windage Plates Assembled on MS9001FA Turbine Flange – Typical Layout

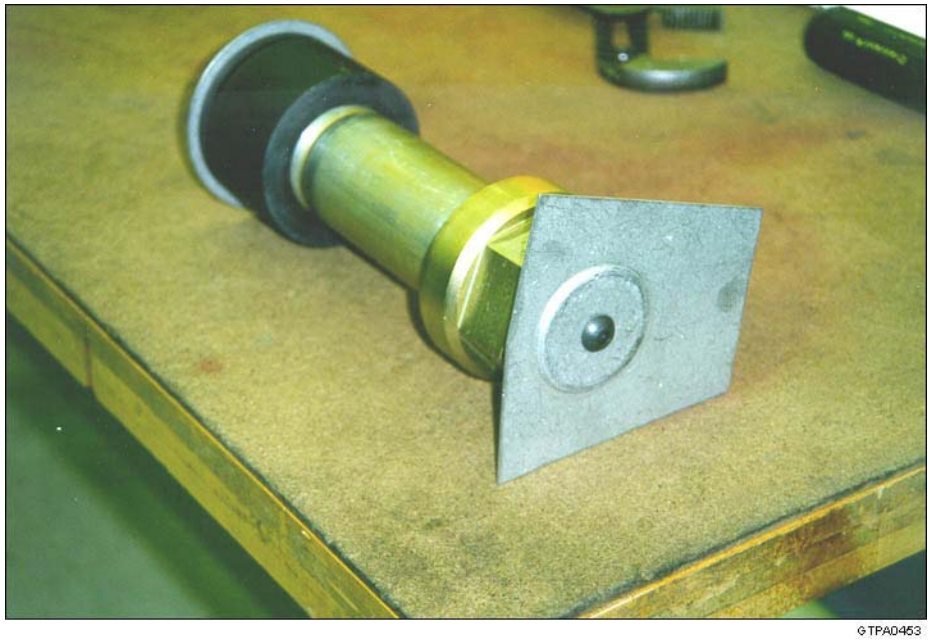


Figure 14. Windage Cover – Coupling End



Figure 15. Windage Cover – Generator or Turbine End

EXHIBIT "A"

Section 1 Addendum to EM Instruction covering MS6001B and MS5001P Gas Turbines with Tension Studs and Nuts.

Section 2 Addendum to EM Instruction covering MS6001FA Gas Turbines to 7A6 or 9A4 Generators with Tension Studs and Nuts.

Section 3 Addendum to EM Instruction covering MS7001EA Gas Turbines to 7A6 Generators or Mechanical Drive Equipment with Tension Studs and Nuts.

Section 4 Addendum to EM Instruction covering MS9001FA Gas Turbines to 324 Generators with Tension Studs and Nuts.

EXHIBIT "A" – Section 1

Addendum to Instruction Manual IM100
For

MS6001B and MS5001P Gas Turbines
with Tension Studs and Nuts

12.0 HAND TOOLS

The following hand wrenches and micrometers will be needed to install the studs.

MS6001B and MS5001P Gas Turbines	
5/8 in. Wrench	5 in. to 6 in. Micrometer
15/16 in. Wrench	6 in. to 7 in. Micrometer
1.0 in. Wrench	10 in. to 11 in. Micrometer

13.0 SPECIAL TOOLS

- Hydraulic Pressure Kit – MP-0130 (hand pump), or AP-0532 (air-driven pump recommended).
- 1-1/8 in. Hydraulic Tensioner Kit – HT-0245, HT-0141 (Zurn load coupling), HT-0251 or HT-0406.
- 1-1/4 in. Hydraulic Tensioner Kit – HT-0176, HT-0219.
- 1-1/2 in. Hydraulic Tensioner Kit – HT-0208.

Note: Refer to Items 1.0 through 5.3 (pages 1 through 8) for a step by step procedure for MS7001FA turbines similar to that used for MS6001B and MS5001P turbines.

14.0 BLEEDING THE SYSTEM

Bleeding the air from the hydraulic system must be performed before stud tensioning begins. Bleed the system as follows.

1. Mount the tensioner on a stud per assembly instructions of Section 6.0.
2. Ensure the tensioner assembly is situated above the pump. The tensioner assembly has two ports, one for pressurizing the system and one for bleeding the system.

3. The bleed port must always be oriented in the uppermost position. The puller tool is shipped with a 5/8 in. hex cone stem bleeder fitting installed.

4. With this fitting loosened, stroke the pump repeatedly until the stream of oil exiting the tool is free of air. Then retighten the fitting.

15.0 ASSEMBLY OF TENSIONER ON STUD

Two types of Tensioner Kits will be encountered in the field. The two types are easily identified by their particular safety cages. One cage is separate from the puller tool while the other is internal. Other design differences and operational characteristics are defined in the following Sections. The two assemblies are depicted in Figures 17 and 18.

All tensioning (pulling) is to be performed from the tapered thread end of the stud with orientation of the stud to the flange as shown in Figures 19, 20 and 21.

15.1 Assembly of Tensioning Kit with Separate Safety Cage

Refer to Tensioner Assembly drawing and Figure 18 for mounting tensioner to flange. Assemble as follows.

1. Open the hydraulic return valve on the pump to allow hydraulic fluid to be pushed back from the puller tool into the reservoir as the puller tool is tightened.

2. Place the spanner ring on the puller side nut.

3. Place and hold the puller tool over the end to be tightened.

4. Insert the puller screw through the puller tool and into the tapered thread end of the stud. Hand tighten.

CAUTION: Ensure assembly is not cross threaded.

5. Tighten the puller screw using an open-end wrench and spanner or Allen wrench, depending upon hardware configuration, on the opposite end of the stud. **(DO NOT WRENCH THE NUT.)**

6. At this point the puller screw must be tight in the stud and the tensioner assembly must be free to rotate.

7. If the puller tool is not free to rotate when the puller screw is tight, then either (1) the piston is not fully retracted, or (2) the stud is not properly positioned in the flange.

1. If the piston is not fully retracted, open the valve on the pump and retighten the puller screw.

2. If the stud is not properly positioned in the flange, the nuts must be repositioned so that the stud is shifted slightly more on the puller tool side. This can be done as follows.

   (1) Slightly loosen the puller tool.
   (2) Back the nut opposite the puller tool.
   (3) Tighten the puller screw side nut to take up the slack.
   (4) Retighten the puller screw and check for looseness of tool.

15.2 Assembly of Tensioner Kit with Integral Safety Cage

Refer to Tensioner Assembly drawing and Figure 18 for mounting tensioner to flange. Mount as follows.

1. Open the hydraulic return valve on the pump to allow hydraulic fluid to be pushed back from the puller tool into the pump reservoir.

2. Place the spanner ring on the puller side of the cylindrical nut.

3. Place and hold the puller tool over the end to be tightened.

4. Insert the puller screw through the puller tool into the tapered thread of the stud and tighten.

CAUTION: Ensure assembly is not cross threaded.

5. Tighten the puller screw using Allen wrenches. **Do not** wrench the Hex nut opposite the puller tool.

6. Install the puller nut until it seats snuggly on the piston, then back off 2 flats. This is particularly important for removal because the stud shortens during disassembly and the tensioner may bind.

7. At this point the Tensioner assembly **must be free to rotate.** The puller screw is tight in the stud and the puller nut has been backed off two flats.

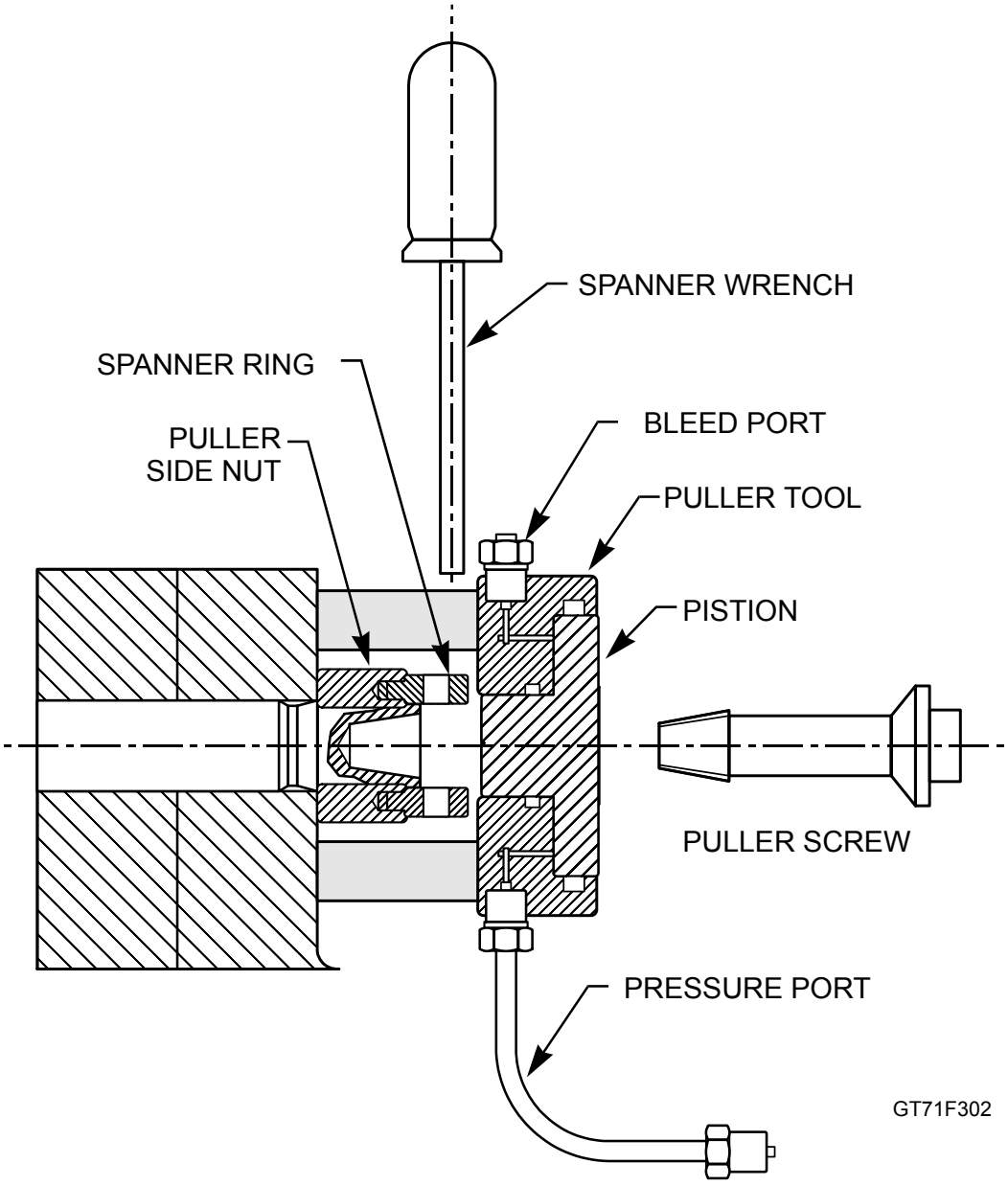


Figure 17. Separate Safety Cage Style Tensioner (Cage Not Shown)

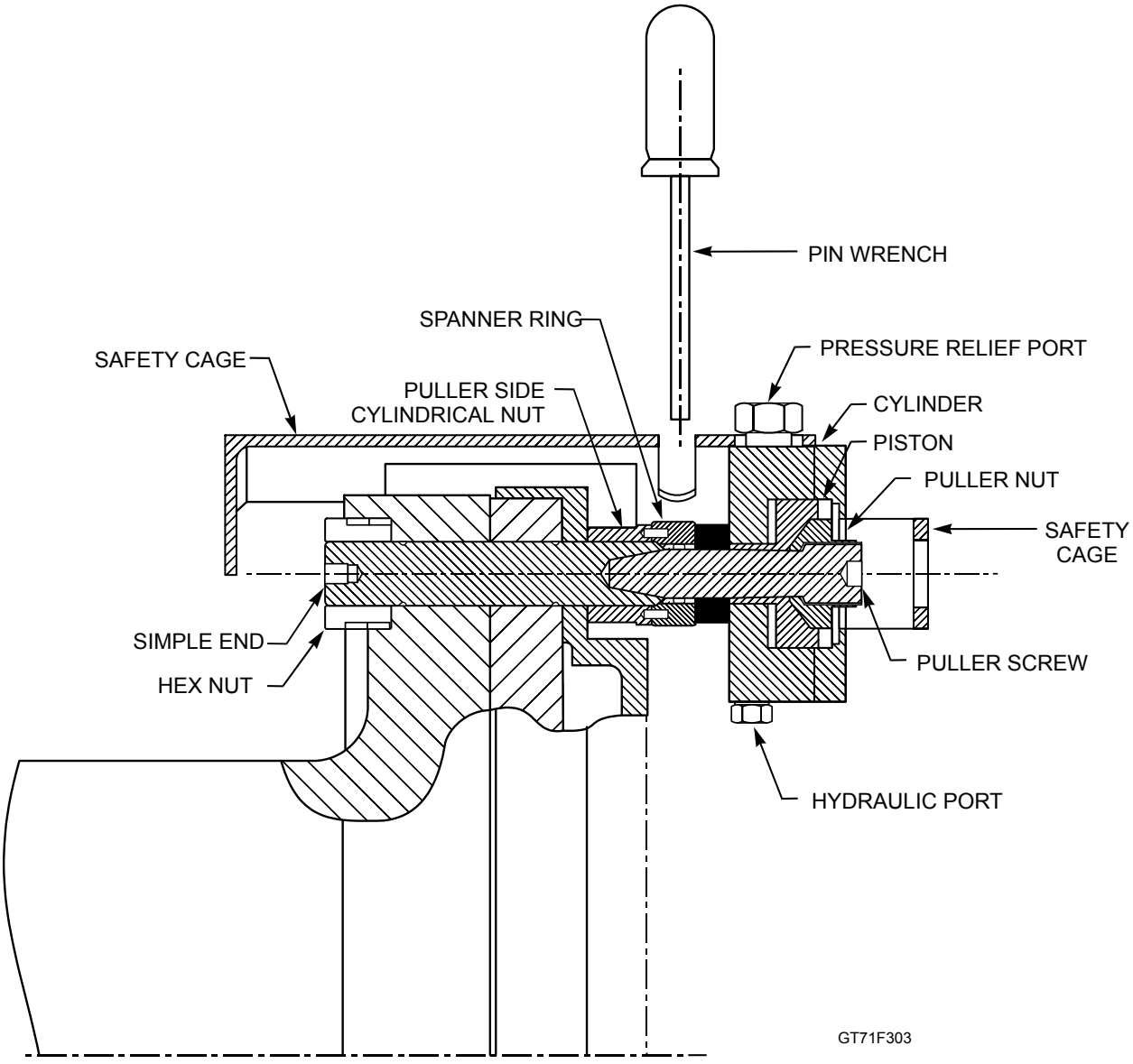


Figure 18. Integral Guard Tensioner Mounted on Turbine Flange

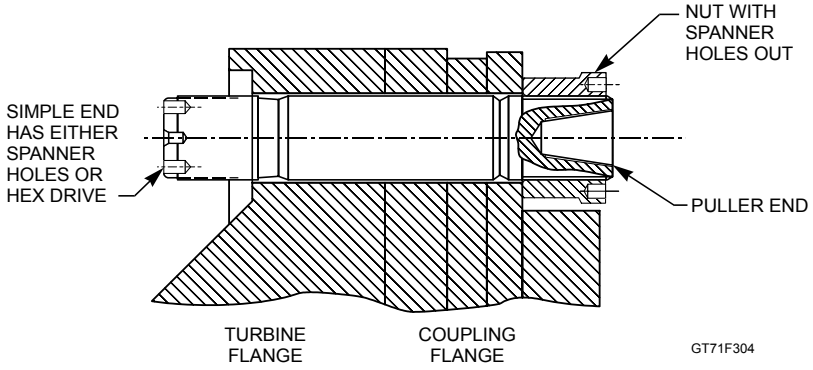


Figure 19. Load Coupling/Load Gear Bolting

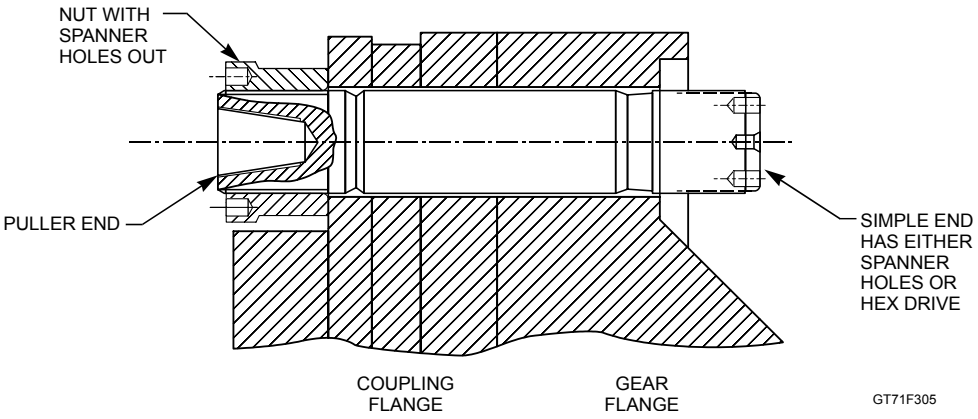


Figure 20. Load Coupling/Load Gear Bolting

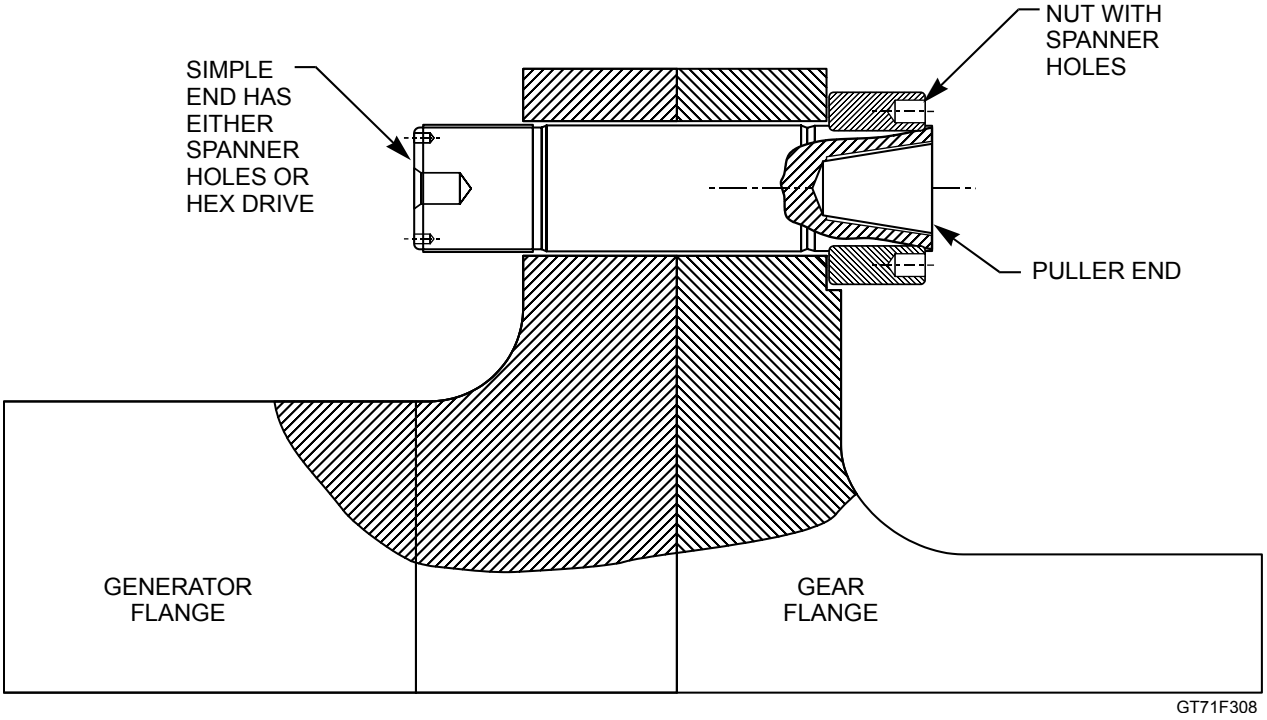


Figure 21. Load Coupling/Generator Hub Bolting



Figure 22. Turbine End View of Stud and Nut

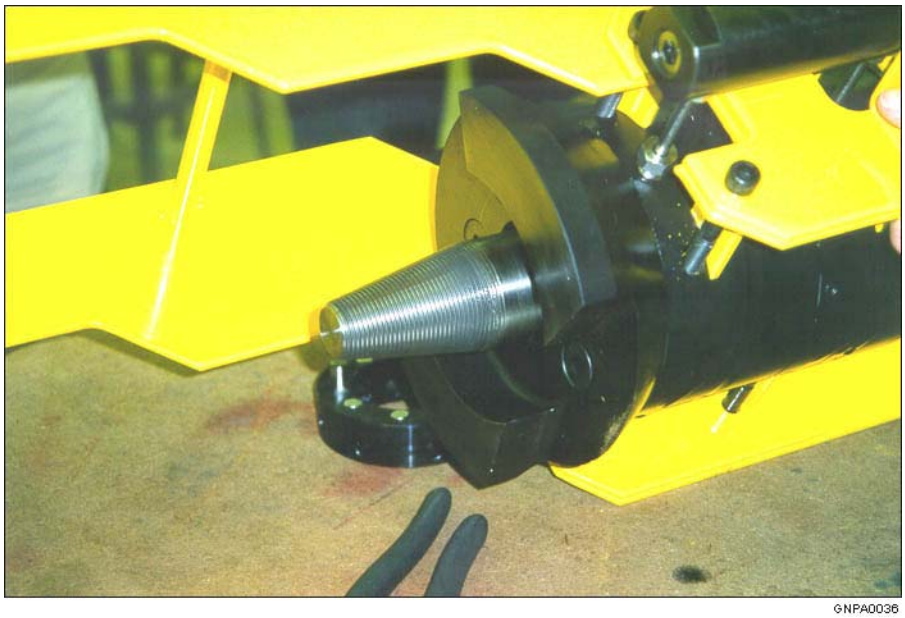


Figure 23. Tensioner Pulling Screw

Note: If the tool does not rotate freely, it is likely that the nuts need to be repositioned so that the stud shifts slightly to the puller side of the flange. See Step 15.1.7.2.

Note: **Do Not** over extend the stud, as this can cause the piston to lose its seal and leak oil.

CAUTION: **Personal injury and equipment damage can occur if the puller screw is not securely engaged with the tapered threads of the stud. Proper engagement is achieved when the puller screw is tight in the stud and the tensioner assembly is free to turn. Refer to Figures 22 and 23.**

WARNING: **To avoid personal injury the safety cage MUST be in place, and hands must be kept out of designated areas at all times when the puller tool is pressurized.**

15.3 Tensioning at 50% Pressure

After the tensioner is installed, apply hydraulic pressure to the tool. Bring the pressure to the 50% level in accordance with the following **MS6001B/MS5001P** table.

MS6001B and MS5001P Gas Turbines		
Flange	**Stud Diameter**	**50% Pressure**
Turbine/Coupling	1.125 in.	9,000 psi
Coupling/Gear	1.125 in.	9,000 psi
Gear/Generator	1.250 in.	6,000 psi
Gear/Generator	1.500 in.	7,500 psi

15.3.1 Tightening the 1.125 Inch Nuts

Hand tighten the cylindrical nuts using the pin wrench and spanner ring. Refer to Figures 17 and 18.

15.3.2 Tightening the 1.250 Inch Nuts

Tighten the cylindrical nuts using the pin wrench and the spanner ring. Refer to Figures 17 and 18. Turn the nut until it bottoms on the flange, then apply torque to turn the nut an additional 10 degrees. This will aid in achieving the desired stretch.

15.3.3 Tightening the 1.500 Inch Nuts

Tighten the cylindrical nut using the pin wrench and the spanner ring. Refer to Section 15.4, "Tensioning at Final Pressure."

15.4 Tensioning at Final Pressure

Repeat the pulling and tensioning procedure at final pressure as described in Section 15.3. Measure the length of the studs after all have been pulled. The required stretch values are listed in the following **MS6001B/MS5001P** table.

CAUTION: DO NOT exceed the pressure marked on the puller. Excessive pressure can damage the stud and the puller screw.

Full Tension – MS6001B and MS5001P Gas Turbines			
Flange	**Size**	**Stretch (Inches)**	**Pressure**
Turbine/Coupling & Coupling/Gear	1.125 in.	.010/.012 in.	18,000 psi
Gear/Generator	1.250 in.	.014/.016 in.	12,500 psi
6A3 Gear/Generator			

15.5 Retensioning

Refer to Section 8.0, "Retensioning."

16.0 THREAD LOCKING

Refer to Section 9.0, "Thread Locking."

17.0 FIELD ASSEMBLY

Refer to Figures 24, 25, 26 and 27.



Figure 24. Top View of Gear to Generator Flange (shows Puller Tool with Safety Cage)

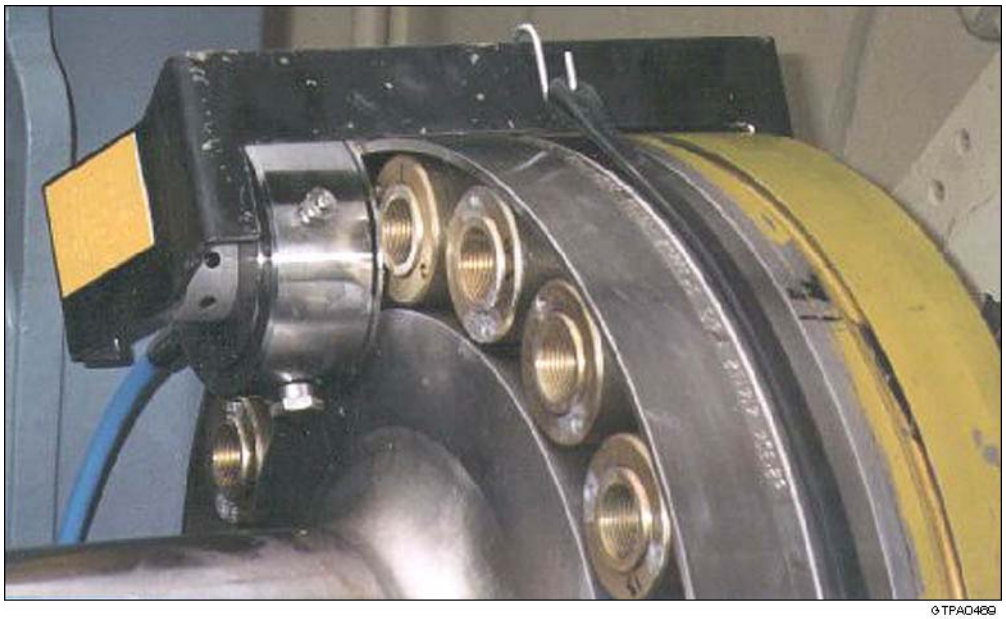


Figure 25. Side View of Puller Tool with Safety Cage

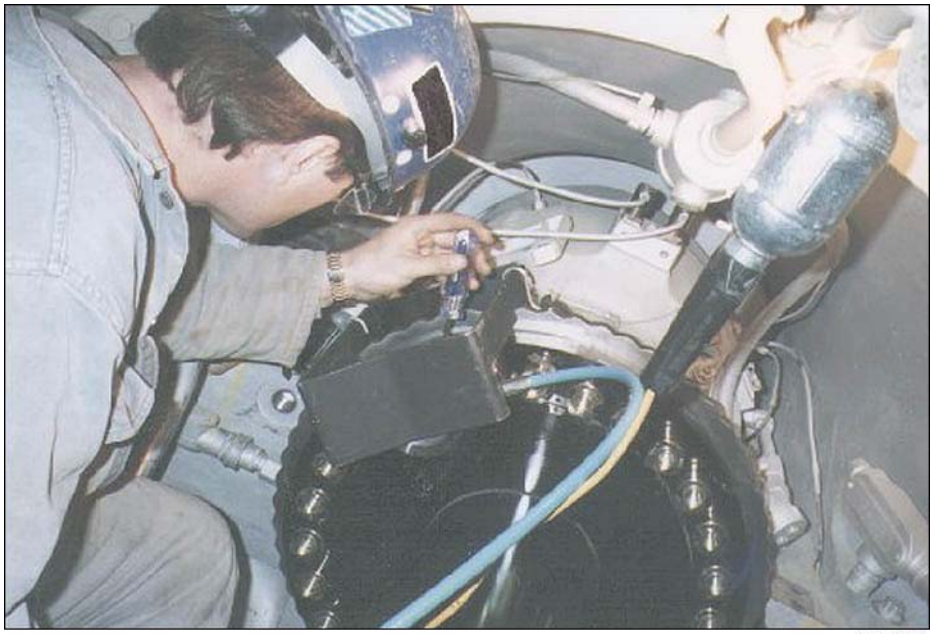


Figure 26. Top View of Puller Tool and Safety Cage Mounted on Coupling to Gear Flange



Figure 27. Side View of Puller Tool and Safety Cage Mounted on Turbine to Coupling Flange

EXHIBIT "A" – Section 2

Addendum to Instruction Manual 1M101
For

MS6001FA Gas Turbines to 7A6 or 9A4 Generators
with Tension Studs and Nuts

18.0 HAND TOOLS

Several hand wrenches and micrometers will be needed to install the studs.

MS6001FA Gas Turbines	
5/8 in. Wrench	3 in. Wrench
15/16 in. Wrench	5 in. to 6 in. Micrometer
1.0 in. Wrench	8 in. to 9 in. Micrometer
2-3/4 in. Wrench	9 in. to 10 in. Micrometer

19.0 SPECIAL TOOLS

- Hydraulic Pressure Kit – MP-0130.
- 1-1/8 in. Hydraulic Tensioner Kit – HT-0141, HT-0245, HT-0251, or HT-0406.
- 2 in. Hydraulic Tensioner Kit – HT-0174, HT-0252, HT-0716.
- 2-3/4 in. Hydraulic Tensioner Kit – HT-0739.

Note: Refer to Items 1.0 through 5.3 (pages 1 through 10) for a step by step procedure for MS7001FA turbines similar to that used for MS6001FA turbines.

20.0 TENSIONING AT 50% PRESSURE

After the tensioner is installed, apply hydraulic pressure to the tool. Bring the pressure to the 50% level in accordance with the following **MS6001FA** table.

Half Tension – MS6001FA Gas Turbines		
Flange	**Stud Diameter**	**50% Pressure**
Turbine/Coupling	1.125 in.	9,000 psi
Coupling/Gear	1.125 in.	9,000 psi
Gear/Generator	2.000 in.	9,000 psi
Gear/Generator	2.750 in.	9,000 psi

20.1 Tightening the 1.125 Inch Nuts

Hand tighten the cylindrical nuts using the pin wrench and spanner ring. Refer to Figures 17 and 18.

20.2 Tightening the 2.000 Inch Nuts

Tighten the cylindrical nuts using the pin wrench and the spanner ring. Refer to Figures 17 and 18. Turn the nut until it bottoms on the flange, then apply torque to turn the nut an additional 10 degrees. This will aid in achieving the desired stretch.

20.3 Tightening the 2.750 Inch Nuts

Tighten the cylindrical nut using the pin wrench and the spanner ring. Refer to Section 15.4 "Tensioning at Final Pressure."

20.4 Tensioning at Final Pressure

Repeat the pulling and tensioning procedure at final pressure as described in Section 15.3. Measure the length of the studs after all have been pulled. The required stretch values are listed in the **MS6001K** table.

CAUTION: DO NOT exceed the pressure marked on the puller. Excessive pressure can damage the stud and the puller screw.

Full Tension – MS6001FA Gas Turbines			
Flange	**Size**	**Stretch (Inches)**	**Pressure**
Turbine/Coupling & Coupling/Gear	1.125 in.	.012/.014 in.	18,000 psi
Gear/7A6 Generator	2.000 in.	.012/.014 in.	18,000 psi
Gear/9A4 Generator	2.750 in.	.009/.011 in.	17,500 psi

20.5 Retensioning

Refer to Section 8.0, "Retensioning."

21.0 THREAD LOCKING

Refer to Section 9.0, "Thread Locking."

22.0 FIELD ASSEMBLY

Refer to Figures 24, 25, 26 and 27.

EXHIBIT "A" – Section 3

Addendum to Instruction Manual 1M110
For

MS7001EA Gas Turbines to 7A6 Generators
or Mechanical Drive Equipment
with Tension Studs and Nuts

23.0 HAND TOOLS

Several hand wrenches will be needed to install the studs.

MS7001EA Gas Turbines	
5/8 in. Wrench	1-1/8 in. Wrench
15/16 in. Wrench	8 in. to 9 in. Micrometer
1.0 in. Wrench	3/8 in. Hex Key Wrench

24.0 SPECIAL TOOLS

- Hydraulic Pressure Kit – MP-0130.
- 2 in. Hydraulic Tensioner Kit – HT-0252, HT-0716.

Note: Refer to Items 1.0 through 5.3 (pages 1 through 10) for a step by step procedure for MS7001FA turbines similar to that used for MS7001E turbines.

25.0 TURBINE TO GENERATOR

25.1 Tensioning at 50% Pressure

After the tensioner is installed, apply hydraulic pressure to the tool. Bring pressure to the 50% level in accordance with the following **MS7001EA** table.

MS7001EA Gas Turbines		
Flange	**Stud Diameter**	**50% Pressure**
Turbine/Coupling	2.000 in.	9,000 psi
Coupling/Generator	2.000 in.	7,000 psi

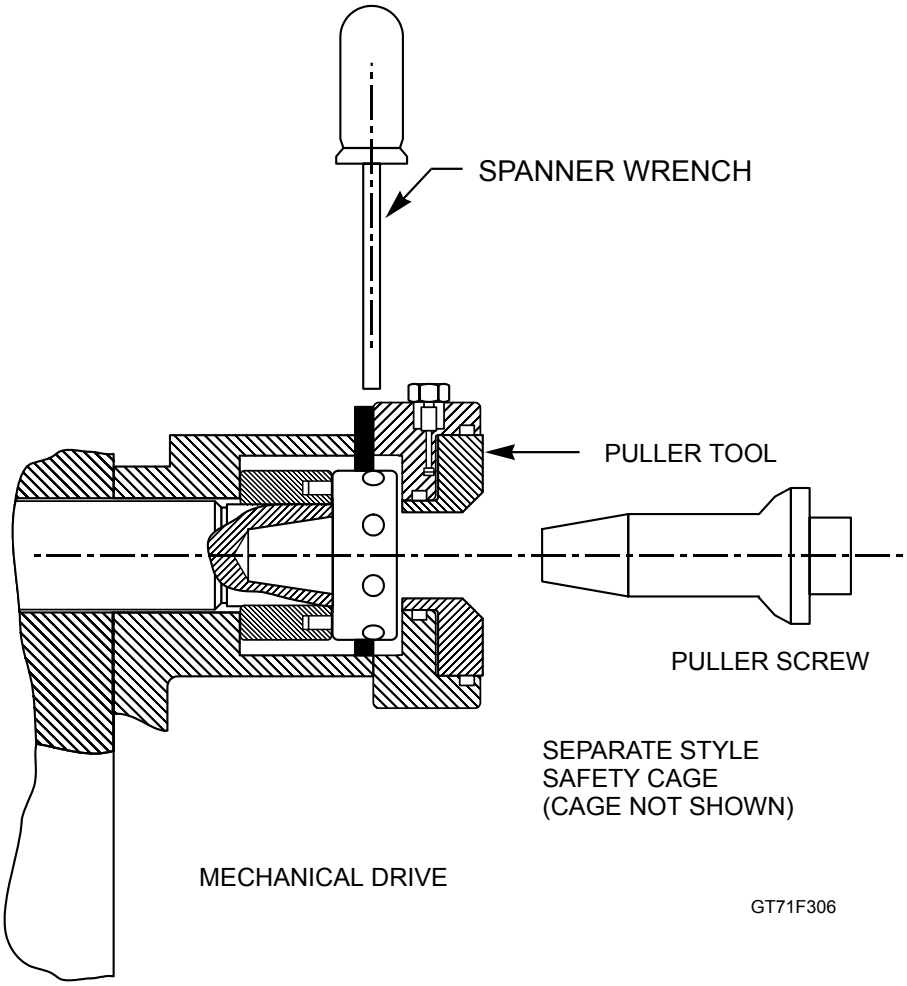


Figure 28. Hand Tighten Cylindrical Nut Using Pin Wrench and Spanner Ring

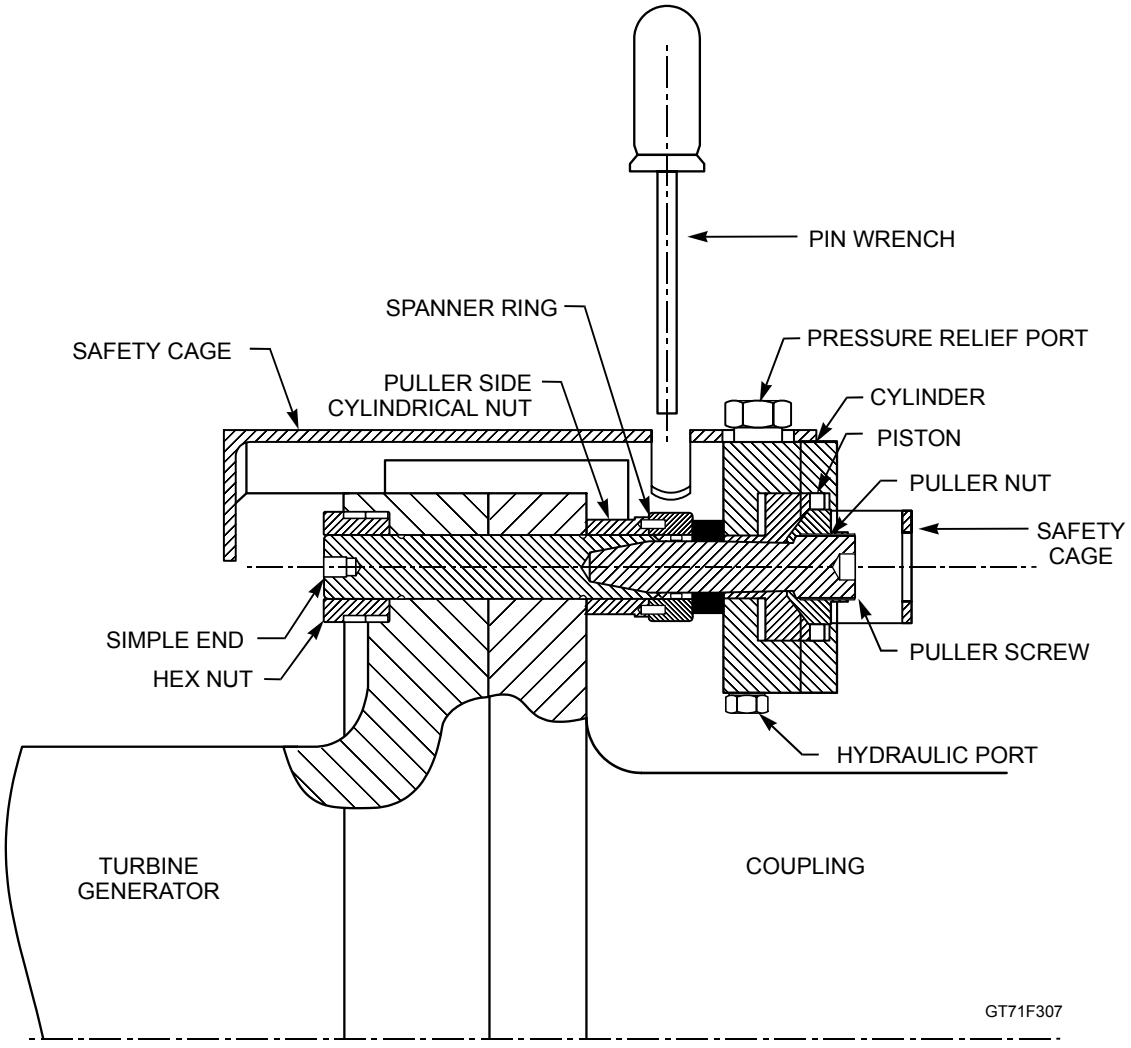


Figure 29. Tightening Spanner Ring

25.2 Tightening of Turbine to Coupling Nuts

Hand tighten the cylindrical nuts using the pin wrench and spanner ring as shown in either Figure 28 or Figure 29. Turn the nut until it bottoms on the flange, then apply torque to turn the nut an additional 10 degrees. This will aid in achieving the desired stretch.

25.3 Tightening Coupling to Generator Nuts

Hand tighten the cylindrical nuts using the pin wrench and spanner ring as shown in Figures 28 and 29.

25.4 Tensioning at Final Pressure

Repeat the pulling and tensioning procedure at final pressure as described in Sections 25.0 and 25.1. Measure the length of the studs after all have been pulled. The required stretch values and final pressure are listed in the following table.

CAUTION: DO NOT exceed pressure marked on the puller. Excessive pressure can damage the stud and the puller screw.

Full Tension – MS7001EA Gas Turbines			
Flange	**Size**	**Stretch (Inches)**	**Pressure**
Turbine/Coupling	2.000 in.	.013/.015 in.	19,000 psi
Coupling/Generator	2.000 in.	.010/.012 in.	15,000 psi

26.0 TURBINE TO MECHANICAL DRIVE

26.1 Tensioning at 50% Pressure

Repeat Section 25.0, Steps 1 through 4.

27.0 TENSIONING AT FINAL PRESSURE

Repeat the pulling and tensioning procedure at final pressure as described in Sections 25.0 and 25.1. Measure the length of the studs after all have been pulled. The required stretch values and final pressure are listed in the following table.

Full Tension - Turbine to Mechanical Drive – MS7001EA Gas Turbines			
Flange	**Size**	**Stretch (Inches)**	**Pressure**
Turbine/Coupling	2.000 in.	.0125/.0145 in.	18,000 psi

28.0 RETENSIONING

Referencing Sections 25.2 and 25.3, correct excessive stretch variations or low stretch values by retensioning all or selected studs to the pressure values in the above table. Final stretch values are to be approved by the lead Installation Technical Advisor.

29.0 FIELD ASSEMBLY

Refer to Figures 24, 25, 26 and 27.

EXHIBIT "A" – Section 4

Addendum to Instruction Manual 1M114
For

MS9001FA Gas Turbines to 324 Generators
with Tension Studs and Nuts

30.0 HAND TOOLS

Several hand wrenches will be needed to install the studs.

MS9001FA Gas Turbines	
5/8 in. Wrench	Impact Wrench
1.0 in. Wrench	1/2 in. Hex Key Wrench
1.0 in. Socket	12 in. to 13 in. Micrometer

31.0 SPECIAL TOOLS

- Hydraulic Pressure Kit – MP-0130 or AP-0532.
- Hydraulic Tensioner Kit – HT-0445.

Note: Refer to Items 1.0 through 5.3 (pages 1 through 10) for a step by step procedure for MS7001FA turbines similar to that used for MS9001FA turbines.

32.0 TURBINE TO GENERATOR

32.1 Tensioning at 50% Pressure

After the tensioner is installed, apply hydraulic pressure to the tool. Bring the pressure to the 50% level in accordance with the following **MS9001FA** table.

CAUTION: DO NOT exceed the pressure marked on the puller. Excessive pressure can damage the stud and the puller screw.

Half Tension – MS9001FA Gas Turbines		
Flange	**Stud Diameter**	**50% Pressure**
Turbine/Coupling	2.750 in.	10,000 psi
Coupling/Generator	2.750 in.	10,000 psi

32.2 Tightening Turbine/Coupling and Coupling/Generator Nuts

Hand tighten the cylindrical nuts using the pin wrench and spanner ring as shown in Figure 7. Turn the nut until it bottoms on the flange.

33.0 TENSIONING AT FINAL PRESSURE

Repeat the pulling and tensioning procedure at final pressure as described in Section 25.2. Measure the length of the studs after all have been pulled. The required stretch values and the final stretch pressure is listed in the following table.

CAUTION: DO NOT exceed the pressure marked on the puller. Excessive pressure can damage the stud and the puller screw.

Full Tension – MS9001FA Gas Turbines			
Flange	**Size**	**Stretch (Inches)**	**Pressure**
Turbine/Coupling	2.750 in.	.022/.024 in.	17,000 psi
Coupling/Generator	2.750 in.	.025/.027 in.	19,000 psi

34.0 FIELD ASSEMBLY

Refer to Figures 3, 4, 5, 8 and 9.

35.0 FIGURES

Figures 30 through 38 show stud, nuts and the hydraulic power unit in shipping cases at the Riverhawk Factory.



Figure 30. MS7001FA Coupling Studs and Nuts in Shipping Case



Figure 31. MS7001FA Coupling Studs and Nuts in Shipping Case



GTPA0457

Figure 32. MS7001FA Coupling Nuts in Shipping Case



GTPA0458

Figure 33. Riverhawk Hydraulic Power Unit

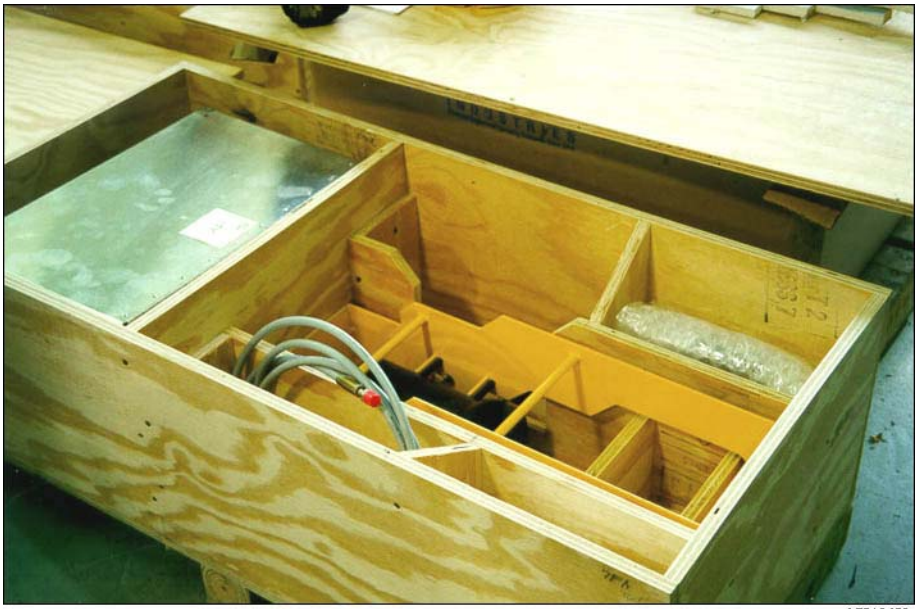


Figure 34. Hydraulic Power Unit and Torque Tensioner Tool in Shipping Case



Figure 35. Hydraulic Power Unit and Torque Tensioner Tool in Shipping Case



Figure 36. Hydraulic Power Unit in Shipping Case

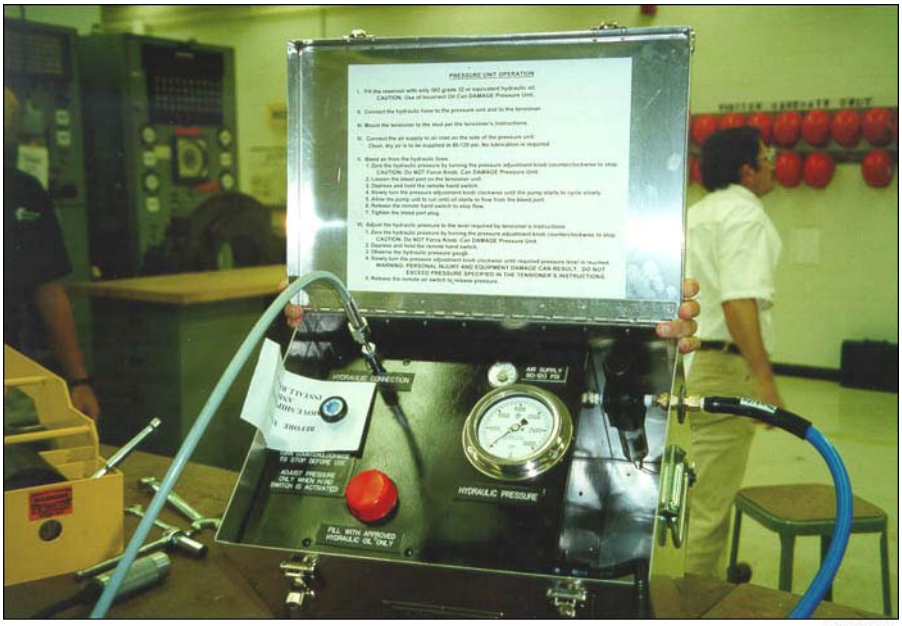


Figure 37. Hydraulic Power Unit on Bench in Riverhawk Factory



Figure 38. MS6001FA Stud and Nut Kit in Shipping Case